

Facing Differences of Similarity: Intra- and Inter-Correlation Unsupervised Learning for Chest X-Ray Anomaly Detection

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Abstract—Anomaly detection can significantly aid doctors in interpreting chest X-rays. The commonly used strategy involves utilizing the pre-trained network to extract features from normal data to establish feature representations. However, when a pre-trained network is applied to more detailed X-rays, differences of similarity can limit the robustness of these feature representations. Therefore, we propose an intra- and inter-correlation learning framework for chest X-ray anomaly detection. Firstly, to better leverage the similar anatomical structure information in chest X-rays, we introduce the Anatomical-Feature Pyramid Fusion Module for feature fusion. This module aims to obtain fusion features with both local details and global contextual information. These fusion features are initialized by a trainable feature mapper and stored in a feature bank to serve as centers for learning. Furthermore, to Facing Differences of Similarity (FDS) introduced by the pre-trained network, we propose an intra- and inter-correlation learning strategy: 1) We use intra-correlation learning to establish intra-correlation between mapped features of individual images and semantic centers, thereby initially discovering lesions; 2) We employ inter-correlation learning to establish inter-correlation between mapped features of different images, further mitigating the differences of similarity introduced by the pre-trained network, and achieving effective detection results even in diverse chest disease

environments. Finally, a comparison with 18 state-of-the-art methods on three datasets demonstrates the superiority and effectiveness of the proposed method across various scenarios.

Index Terms—Medical anomaly detection, correlation learning, feature fusion, transfer learning, chest X-ray.

I. INTRODUCTION

X-RAY images are commonly used by physicians to diagnose and evaluate various medical conditions, such as pneumonia, chest fractures, and pneumothorax. Despite being considered intuitive, accurate, and an efficient diagnostic tool, X-ray images still require interpretation by skilled radiologists. Even experienced practitioners, who rely on visual inspection of chest X-ray images, face challenges such as time-consuming analysis and the potential for missed diagnoses and misdiagnoses. Several deep learning methods have shown significant efficacy in automatically analyzing X-ray images, thus easing the workload burden on radiologists [1], [2], [3], [4], [5], [6]. However, the scarcity of large, well-annotated X-ray datasets impedes supervised learning performance. Therefore, unsupervised anomaly detection has become a primary research focus in X-ray medical imaging [7], [8], [9], [10], [11], [12], [13], [14]. However, compared to natural datasets, the differences between normal and abnormal data in X-ray images are more subtle. Therefore, this poses significant challenges for chest X-ray anomaly detection.

Currently, anomaly detection models solely utilize normal data to construct feature representations. During the testing phase, these models identify anomalies by analyzing whether the feature representations significantly deviate from the normal distribution. Most medical anomaly detection [7], [8], [9], [10] relies on simulating pseudo-anomalous data to construct effective feature representations. Due to the diverse and unpredictable nature of anomalies, simulating pseudo-anomalous data does not closely match real anomalies. Moreover, simulating an anomaly set containing all possible anomalies is impractical [15]. Methods based on Teacher-Student architectures [12], [16], [17], [18], [19], [20] implicitly describe the normal feature distribution by optimizing the differences between teacher and student features distribution (learning normal feature distribution). However, the student network may unexpectedly produce abnormal feature distribution similar to

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those in the teacher network. To overcome this issue, methods based on distribution mapping (mapping the normal feature distribution to a specific distribution) have shown improved potential. Recently, distribution mapping methods primarily employ techniques such as normalization flows [21], [22], [23] and feature fine-tuning [13], [14], [24], [25], [26], [27]. The normalization flow-based methods primarily transform the output of the pre-trained network into a Gaussian distribution. Feature fine-tuning methods primarily map the output of the pre-trained network to the feature distribution in a memory bank or reduce the feature distribution differences between various positive samples through contrastive learning strategies. The pre-trained network plays a crucial role in distribution mapping methods and is also essential for the field of medical anomaly detection [28]. However, the pre-trained network is trained on large-scale datasets like ImageNet [29], which do not include medical images. Although these methods perform well on datasets such as MVTecAD [30], their application to more subtle X-ray datasets may be limited by the introduction of bias [31], [32]. These biases stem from differences of similarity between the pre-trained dataset domain and the target dataset domain. For example, in the ImageNet, the structure of objects and scenes is conveyed through a combination of color, shape, and texture, which are highly complex and variable. In contrast, structures in X-rays are typically more direct and apparent because X-rays primarily depict the internal anatomical structures of the body in a straightforward and intuitive manner, focusing mainly on internal rather than external features of objects.

The motivation here is to address the bias introduced by using the pre-trained network, while detecting subtle lesions—from the way humans discover anomalies. When an ordinary person without medical knowledge detects lesions on chest X-rays, apart from obvious lesions (**intra-correlation learning**), the most effective method is to search for lesions by pairwise comparison of the same positions (**inter-correlation learning**). Inspired by this, we propose an intra- and inter-correlation learning framework for anomaly detection in chest X-rays, aimed at **Facing Differences of Similarity (FDS)**. Firstly, we consider that X-ray images usually focus on specific body regions and therefore exhibit similar structural features. To fully utilize this fixed anatomical structure information, we propose an Anatomical-Feature Pyramid Fusion (AFPF) module for feature fusion. The features fused by the AFPF module are initialized by a trainable feature mapper and stored in a feature bank as semantic centers. Next, we reevaluate the relationship between mapped features and semantic centers, identifying two mapping scenarios that influence the detection of subtle lesions: fuzzy mapping and error mapping. Fuzzy mapping refers to mapping features close to multiple related semantic centers. Error mapping involves mapping features near semantic centers that are semantically mismatched. Therefore, we propose intra- and inter-correlation learning to optimize the mapping ability of the feature mapper, enabling it to detect subtle lesions. Specifically, intra-correlation learning establishes intra-correlation between individual images and semantic centers. In the face of fuzzy mapping, intra-correlation is established to determine the unique center

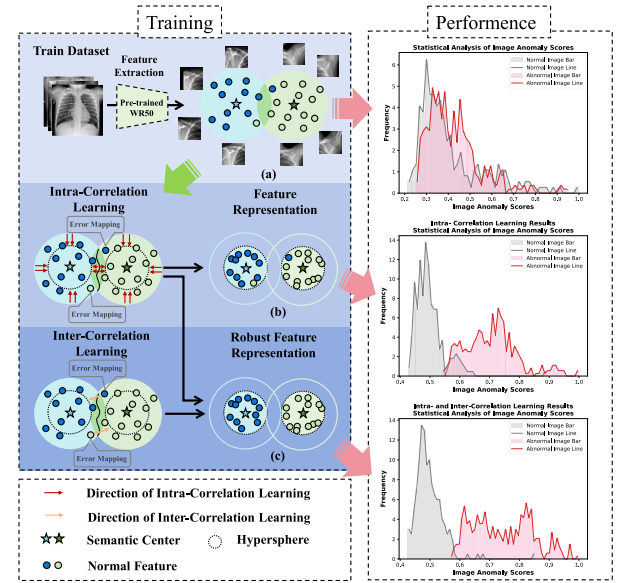


Fig. 1. (a) The use of a network pre-trained on ImageNet has led to differences of similarity, thereby causing significant differences in feature distribution. (b) Through intra-correlation learning, features are mapped into hyperspheres corresponding to their semantics, thereby effectively constructing feature representations. (c) By integrating inter-correlation learning, error mapping caused by differences of similarity are corrected, thereby alleviating significant differences in feature distribution. (Note: Features of various colors represent distinct semantics.)

of fuzzy mapping features, resulting in a higher-density representation of mapped features. Inter-correlation learning establishes inter-correlation between different images. In the case of error mapping, the mutual antagonism of intra- and inter-correlation learning guides the optimization direction of error mapping, as shown in Fig. 1. Finally, the feature mapper trained by the intra- and inter-correlation learning framework can be used for anomaly detection. Our FDS is compared with several state-of-the-art methods on the pediatric pneumonia ZhangLab dataset, the adult pneumonia RSNA dataset, and the multi-disease CheXpert dataset. The results show that our proposed method outperforms other methods in anomaly detection performance. The main contributions are as follows:

- We proposed an intra- and inter-correlation learning framework FDS for chest X-ray anomaly detection, which applies Facing the Differences of Similarity to mitigate the biases introduced by the pre-trained network. This framework is able to construct more robust feature representations, thereby demonstrating exceptional capabilities in anomaly detection.
- We have developed an AFPF module specifically for X-ray images, designed to preserve local details while enhancing its effectiveness in capturing subtle yet large-scale lesion patterns, including patchy, cloudy, or filamentous appearances.
- We conducted comprehensive experiments using three distinct datasets: the pediatric pneumonia ZhangLab dataset, the adult pneumonia RSNA dataset, and the multi-disease CheXpert dataset. We compared FDS with several state-of-the-art baseline methods. Results from these experiments showed that our FDS consistently outperforms other methods in detecting anomalies across

pediatric and adult populations, as well as under various disease conditions.

The structure of this paper is organized as follows: Section II discusses the application of anomaly detection in natural and medical images. Section III offers an in-depth explanation of our proposed FDS framework for chest X-ray anomaly detection. Section IV presents extensive experimental results to validate the effectiveness of FDS. Section V discusses the advantages and disadvantages of FDS. Finally, Section VI provides a summary of our work.

II. RELATED WORK

A. Anomaly Simulation-Based Anomaly Detection

While there are no abnormal data in the training dataset for anomaly detection, it is possible to simulate pseudo-anomalous data, thereby training the anomaly detection model in self-supervised learning. A simple and effective strategy, known as CutPaste [33], randomly selected a rectangular region in the original image and then copied and pasted the content to different locations in the image.

In the field of medical anomaly detection, Tan et al. successfully utilized foreign patch interpolation (FPI) [7] and Poisson image interpolation (PII) [8] to simulate anomalies based on natural variations in normal anatomy, particularly in abnormal images of brain MRI and abdominal CT scans. NSA [9] also explored simulating more natural synthetic anomalies in X-ray images through the integration of Poisson image editing and parameter adjustments for anomaly localization. Anat-Paste [34] introduced an anatomical-aware pasting method that uses a threshold-based lung segmentation preprocessing task to create pseudo-anomalies in normal chest X-ray. Due to the subtle differences in medical images compared to natural datasets, [10] proposed CropMixPaste to generate enhanced predictions of density shadow-like anomalies and encouraged the model to detect local irregularities. However, the occurrence of synthetic anomalies does not closely match real anomalies. In practice, simulating an anomaly set that includes all possible lesions is impossible due to the diverse and unpredictable nature of lesions [15].

B. Teacher-Student Architecture-Based Anomaly Detection

The Teacher-Student architecture is based on a pre-trained teacher network and a trainable student network, accurately localizing by leveraging the expression differences between the teacher and student networks. Various solutions have been proposed in the past to enhance the discrimination and localization capabilities of different anomaly phenomena. For instance, [17] and [18] designed decoder architectures for the student network, avoiding the drawbacks of having identical architectures and data flows between teacher and student networks. Reference [19] introduced simulated anomalies to enhance the constraints on anomalous data. Simultaneously, [19] employed a segmentation network to adaptively fuse multi-level feature similarities, aiming for more accurate anomaly localization. Reference [20] mitigated the limitations of Teacher-Student architecture by compressing the latent space and simulating pseudo-anomalous data to suppress anomalous features.

In the field of medical anomaly detection, [11] introduced a contrastive learning paradigm to address the issue of collapse in Teacher-Student architecture. Reference [12] achieved accurate classification of X-ray images by dynamically maintaining a memory queue based on spatial position information to suppress abnormal features. However, when attempting to minimize the differences between normal feature distributions, the student network may unexpectedly generalize and produce abnormal feature distributions similar to those in the teacher network [19].

C. Distribution Mapping-Based Anomaly Detection

Methods based on distribution mapping utilize a pre-trained network to extract embedding vectors from normal training images, subsequently mapping these embedding vectors into a specific distribution. Distribution mapping is primarily achieved through techniques such as normalization flow [21], [22], [23] and feature fine-tuning [13], [14], [24], [25], [26], [27]. During the testing phase, anomaly scores are determined by calculating the distance between the test images and the feature distribution. Csflo [21] and Fast-flow [22] suggested transforming normal feature distributions into Gaussian distributions through normalization flow [35]. Padim [24] performed feature extraction on sample patches, estimated the Gaussian distribution of patches, and calculates anomaly scores using Mahalanobis distance. On the other hand, PatchCore [25] constructed a feature bank with the most representative patches to maximize normality. CFA [26] adopted feature adaptation based on a coupled hypersphere layer, enhancing the distinguishability of subtle anomaly. Reference [27] addressed the collapse issue that occurs within the unit space with standard contrastive loss by proposing the Mean-Shifted contrastive loss.

In the field of medical anomaly detection, there is relatively less exploration of methods based on distribution mapping. Reference [13] employed differentiable top-k feature adaptation to learn an attention-enhanced patch descriptor, improving the ability to detect anomalies in complex medical imaging data. Reference [14] proposed SubMSF to address the challenges of model collapse or feature deterioration that may occur in the abnormal detection of leukocyte medical images. However, techniques relying on distribution mapping utilize extensive datasets such as ImageNet [29] for pre-training. While they demonstrate effectiveness on industrial datasets like MVTecAD [30], their applicability to medical data is constrained, potentially introducing biases in medical anomaly detection [31], [32]. Reference [36] attempted to circumvent pre-trained network by employing defect contrast positioning but did not observe notable performance enhancements in detecting subtle anomalies in X-ray images. Consequently, further investigation is warranted to refine the distribution mapping method for analyzing chest X-ray images.

III. METHOD

A. Overall Framework

In this section, we will introduce the proposed anomaly detection method FDS for chest X-ray images based on

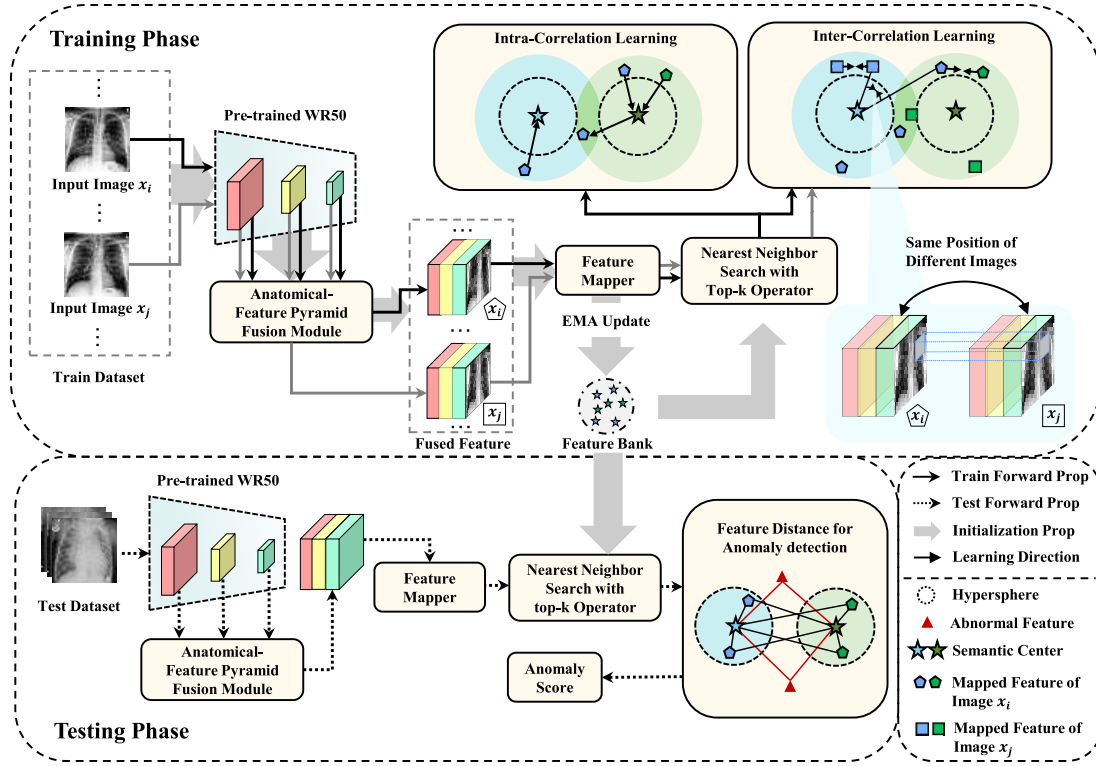


Fig. 2. Overview of the proposed framework FDS: Initially, we extract features using a pre-trained network and employ an Anatomical-Feature Pyramid Fusion (AFPF) module to fully leverage similar structures in X-ray images for obtaining fusion features. Subsequently, these fusion features are initialized by a trainable feature mapper and stored in a feature bank. Next, intra- and inter-correlation learning are utilized to search the feature bank, thereby guiding the construction of robust feature representations by the feature mapper. During testing, we compute the Euclidean distance between the mapped features and the feature bank center to derive the anomaly score. (Note: Features of various colors represent distinct semantics, and examples x_i and x_j are from the same batch.)

intra- and inter-correlation learning, as shown in Fig. 2. Reiss et al.'s research [37] suggests that using a pre-trained network can output relevant features for anomaly detection. Therefore, we use the WideResNet50 model pre-trained on ImageNet as the initial feature extractor σ . The initial training dataset X_{train} consists of normal images ($\forall x_i \in X_{train} : y_i = 0$), where $y_i \in \{0, 1\}$ represents whether the image x_i is normal (0) or abnormal (1). The testing dataset X_{test} includes both normal and abnormal images ($\forall x_i \in X_{test} : y_i \in \{0, 1\}$). We set the output of the j -th layer as $\sigma_j(x_i)$, we use the output of the first three blocks ($j \in \{1, 2, 3\}$) of the feature extractor σ as the features for subsequent fusion. We use the Anatomical-Feature Pyramid Fusion Module $AFPF(\cdot, \cdot, \cdot)$ to obtain fused features $F_i \in R^{T \times D}$, where $T = HW$, H and W are the height and width of fused features, and D represents the sum of dimensions of the feature extractor σ . Current anomaly detection methods [13], [26] use a network pre-trained on the ImageNet dataset to extract features. By reducing the distance from features to their nearest semantic center, it achieves high-density clustering of features around the semantic center. Therefore, anomaly detection can be performed based on the degree to which features deviate from their semantic center. However, it is noteworthy that because there is a semantic difference between the ImageNet dataset and the target chest X-ray dataset, these nearest semantic centers do not necessarily represent semantically similar semantic centers. Therefore, using a network

pre-trained on the ImageNet dataset may lead to biases in feature mapping. In response to this issue, we have conducted a detailed analysis of the mapping relationship between features and semantic centers, identifying the following three main mapping scenarios: (1) Correct mapping: The feature is accurately mapped near a semantic center that matches similar semantics. (2) Fuzzy mapping: The feature is mapped near multiple semantic centers, leading to unclear semantic expressions. (3) Error mapping: The feature is mapped near a semantic center that does not match its semantics. The specific mapping scenarios are shown in Fig. 3. We employ intra- and inter-correlation learning to implement distinct coping strategies tailored to various mapping scenarios. To facilitate this, we use a feature mapper $\phi(\cdot)$ based on the feature bank c to establish the relationship between the mapped feature $\phi(F_{i,t})$ and the semantic center c_t , transferring knowledge from the ImageNet domain to the target domain of chest X-ray images. where, $F_{i,t} \in R^D$, $t \in \{1, \dots, HW\}$, and c_t is the nearest neighbor searched in the feature bank c for the feature $F_{i,t}$. Our objective is to use inter-correlation learning to leverage the information correlation at the same positions in different images to transform error mapping into correct mapping. Intra-correlation learning is employed to establish mutual correlations between the mapped features and the semantic centers. Intra-correlation learning transforms fuzzy mapping into correct mapping and ensures high-density expression of features for correct mapping. Therefore, during the testing

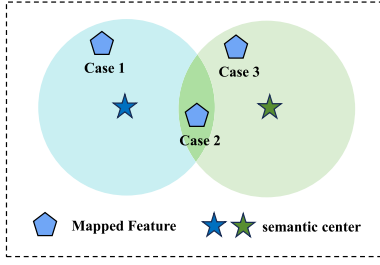


Fig. 3. Differences between the ImageNet domain and the target chest X-rays domain give rise to various mapping scenarios. (**Case 1**) The mapped feature is close to its corresponding semantic center, termed Correct Mapping. (**Case 2**) The mapped feature is equidistant from multiple semantic centers, termed Fuzzy Mapping. (**Case 3**) The mapped feature is close to the semantic center, but contradicts its intended semantic, termed Error Mapping. (Note: Features of various colors represent distinct semantics.)

phase, abnormality is measured by calculating the Euclidean distance between the sample and the semantic centers.

B. Feature Extraction

1) *Anatomical-Feature Pyramid Fusion Module*: The outputs from different layers represent distinct information [38]. The output from lower layers is more characteristic of the structural properties of X-ray images, while the output from higher layers contains more semantic information. Therefore, we use the *AFPF*($\cdot, \cdot, \cdot$) to obtain fused features F_i . This helps the model learn more discriminative features, thereby improving its performance. The *AFPF* module is defined as follows:

$$F_i = AFPF(\sigma_1(x_i), \sigma_2(x_i), \sigma_3(x_i)) \quad (1)$$

Now, let's delve into the specific details of this module to understand how to effectively fuse the outputs of lower and higher-level features to obtain information with both local details and global context. Notably, the ImageNet pre-trained network, which is primarily tailored for natural image classification, introduces a bias that is inappropriate for medical image anomaly detection. Consequently, utilizing the features extracted by the feature extractor σ is not optimal. To address this issue, average pooling is employed to average values within the pooling window, significantly reducing the position sensitivity of local features and ensuring translation invariance [39]. Considering the structural and positional variations in X-ray images, average pooling, denoted $avg(\cdot)$, is applied to $\sigma_j(x_i)$ for $j \in \{1, 2\}$. Furthermore, max pooling, which captures the maximum value within the pooling window, preserves the most significant features and enhances their discriminability [40]. Given the consistent patient postures in X-ray anomaly detection and the primarily semantic distinctions between abnormal and normal images, we augment semantic feature discriminability by extracting local maxima from $\sigma_{j=3}(x_i)$ using max pooling, indicated by $max(\cdot)$. Additionally, the diffuse interstitial pattern of pneumonia is characteristically manifested by reticular opacities, ground-glass opacities, and either fine nodular or patchy infiltrates. To capture these long-range dependencies and enhance the model's sensitivity and generalization to the data, we use Multihead-Attention to aggregate the features obtained from max pooling. Finally, the processed features are upsampled to

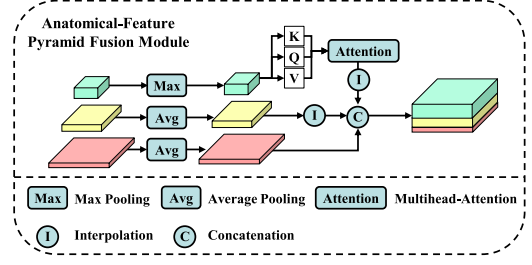


Fig. 4. For similar structures in X-ray images, we propose an *AFPF* module. By combining outputs from different levels of the feature extractor, we obtain more comprehensive features.

the same size and concatenated to obtain the required fused features F_i . The proposed fusion module is shown in Fig. 4 and defined as follows:

$$f_{j,i} = \begin{cases} avg(\sigma_j(x_i)), & j \in \{1, 2\} \\ attn(max(\sigma_j(x_i))), & j \in 3 \end{cases} \quad (2)$$

$$F_i = Con[f_{1,i}, i(f_{2,i}), i(f_{3,i})] \quad (3)$$

where $avg(\cdot)$ represents average pooling, $max(\cdot)$ represents max pooling, $attn(\cdot)$ represents Multihead-Attention, $i(\cdot)$ represents upsampling using bilinear interpolation, and $Con[\cdot, \cdot, \cdot]$ represents Concatenation operation.

2) *Feature Bank*: In order to better utilize the anomaly-free image features extracted by the model, current methods [13], [25], [26], [37] use a feature bank for storage. By building a feature bank, we can better guide the feature mapper $\phi(\cdot)$ for intra-correlation and inter-correlation learning to face differences of similarity. Feature bank is represented as $c \in \mathbb{R}^{T \times D}$, where $T = HW$ is the size of the feature bank, which corresponds to the number of semantic centers. H and W represent the height and width of the fused features, respectively. We dynamically update our feature bank using the Exponential Moving Average (EMA):

$$c_n = \frac{(n-1)c_{n-1} + \phi(F_n)}{n} \quad (4)$$

where n represents the number of updates to the feature bank. After N updates, the feature bank c will contain the overall average semantic features of all data, where N represents the number of images in the training dataset. The formula for the entire process unfolds as follows:

$$c = \frac{1}{N} \sum_i \phi(F_i) \quad (5)$$

C. Intra-Correlation Learning

In the field of medical anomaly detection, pre-trained networks play a crucial role [28]. Most current methods [24], [25], [26] utilize these networks for feature extraction. These pre-trained networks provide generic features (such as texture and edge features) from the ImageNet domain to the target domain, enabling transfer learning capabilities [41]. This generic feature indicates a certain degree of similarity between the ImageNet domain and the target domain. We leverage this similarity to establish associations between mapped features and semantic centers through intra-correlation, effectively handling both correct mapping and fuzzy mapping.

Initially, intra-attraction ensures that during correct mapping, the mapped features exhibit high-density expression around the semantic centers. However, due to differences of similarity between the ImageNet and target domains, mapped features may not anchor to a semantic center, leading to fuzzy mapping. To address this issue, fuzzy mapping is transformed into correct mapping through intra-repulsion using additional hard negative semantic centers.

1) *Intra-Attraction*: To achieve primarily the transfer learning from the pre-trained dataset to the target chest X-ray dataset, we guide the correctly mapped features $\phi(F_{i,t})$ to densely surround the feature bank c . By searching for the K nearest semantic centers $\left\{c_t^k \mid k \in \arg\min_K (D(\phi(F_{i,t}), c_t))\right\}$ in the feature bank c , the mapped feature $\phi(F_{i,t})$ is then mapped into a hypersphere with a radius of R centered at c_t^k . The value of R allows gradient descent, and by continuously adjusting R , the mapped features exhibit a high-density representation. $L_{intra-att}$ is set as a penalty when the distance from c_t^k exceeds the radius R :

$$L_{intra-att} = \frac{1}{TK} \sum_{t=1}^T \sum_{k=1}^K \max \left\{ 0, D(\phi(F_{i,t}), c_t^k) - R \right\} \quad (6)$$

2) *Intra-Repulsion*: In X-ray imaging, we observe the ‘overlap’ phenomenon of various internal tissue structures, which occurs because multiple layers of tissues with different densities are presented simultaneously on the same plane image. This overlapping effect causes some tissue structures to appear similar in the image, thereby leading to fuzzy mapping. To address the situation of fuzzy mapped feature $\phi(F_{i,t})$ simultaneously belonging to multiple semantic centers c_t^k , we employ additional hard-negative semantic centers to repel the fuzzy mapped feature, thereby mapping the fuzzy mapped feature $\phi(F_{i,t})$ to the correct semantic center c_t . By searching for the K -th to $K+J$ -th nearest semantic centers $\left\{c_t^j \mid j \in \arg\min_{[K, K+J]} (D(\phi(F_{i,t}), c_t))\right\}$ from the mapped feature $\phi(F_{i,t})$ to the neighboring semantic center c_t , we then repel the mapped feature $\phi(F_{i,t})$ from a hypersphere centered at c_t^j with a radius of R . This process maps the fuzzy mapped feature $\phi(F_{i,t})$ to the correct c_t^k hypersphere. Therefore, we set $L_{intra-far}$ as the repulsion penalty imposed by c_t^j :

$$L_{intra-rep} = \frac{1}{TJ} \sum_{t=1}^T \sum_{j=1}^J \max \left\{ 0, R^2 - D(\phi(F_{i,t}), c_t^j) - \alpha \right\} \quad (7)$$

The hyperparameter α is used to control the balance between intra-attraction and intra-repulsion.

D. Inter-Correlation Learning

When a feature is mapped to a neighboring semantic center, that feature shares similar semantics with this semantic center. However, the accuracy of mapping is closely related to the quality of features extracted by the pre-trained network. Since

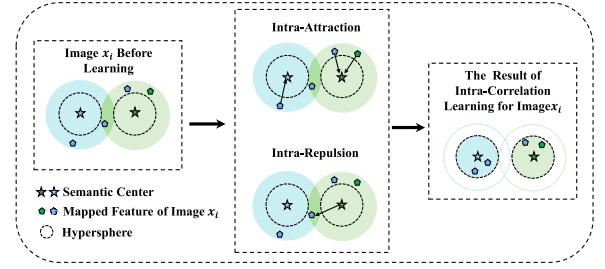


Fig. 5. The optimization process of intra-correlation learning effectively handles fuzzy mapping, ensuring a unique semantic center for fuzzy mapping. However, when features are erroneously mapped, it results in incorrect optimization and shifts the feature to the wrong semantic center. (Note: Features of various colors represent distinct semantics.)

we used a network pre-trained on ImageNet [29], which does not contain medical images, and there are significant differences between the ImageNet domain and the chest X-ray dataset domain [31], [32]. These differences imply that features semantically similar in the ImageNet domain do not necessarily maintain the same semantic similarity in the chest X-ray image domain. For example, in the ImageNet, similar colors or grayscale levels do not consistently reflect similarity, as they might contain completely different objects or scenes. However, in X-ray images, similar gray levels may indicate similar types of tissues or pathological states. These differences of similarity across domains may lead to error mapping, significantly affecting the model’s ability to represent features and further impacting the accuracy of lesion detection. As illustrated in example x_i in Fig. 5, if the mapped feature $\phi(F_{i,t})$ is semantically dissimilar to its neighboring semantic center c_t , then during the intra-correlation learning process, this feature might be erroneously attracted into the hypersphere of the dissimilar neighboring semantic center c_t .

To address this issue, we propose inter-correlation learning to establish relationships between mapped features of different X-ray images. Our aim is to approximately capture the semantic information within the target X-ray image domain by utilizing the inter-correlation between different X-ray images. This approach guides the erroneously mapped features $\phi(F_{i,t})$ towards semantic centers c_t that are semantically similar, thereby enhancing the model’s feature representation capability.

1) *Inter-Attraction*: Since subjects in X-ray images typically maintain a fixed posture, features at the same positions usually have similar semantic information. Therefore, inter-correlation learning selectively reduces the distance between mapped features at the same position in different images, moving the mapped features $\phi(F_{i,t})$ closer to the correct semantic center c_t , thereby mitigating the issue of error mapping. Firstly, to achieve a uniform distribution of the mapped feature $A_{i,t} = \phi(F_{i,t})$ around the semantic center, as opposed to the unit hyperspherical origin o , the distance from the center is updated from the original origin o to the semantic center c_t .

$$C_t \vec{A}_{i,t} = O \vec{A}_{i,t} - \vec{OC}_t \quad (8)$$

where $\vec{C_t A_{i,t}}$ represents the vector from the semantic center c_t to the mapped feature $A_{i,t}$, $O \vec{A}_{i,t}$ represents the vector from

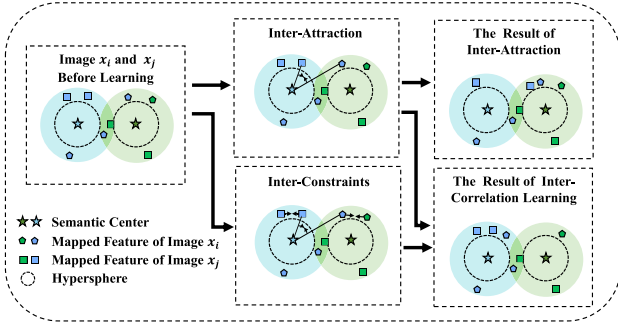


Fig. 6. The Importance of inter-constraints: Sole reliance on inter-attraction does not ensure the correct optimization direction. Through inter-constraints, the mapped features influenced by inter-attraction can adjust their positions based on the distance to neighboring features, thereby avoiding erroneous optimization. (Note: Features of various colors represent distinct semantics, and examples x_i and x_j are from the same batch.)

the origin o to the mapped feature $A_{i,t}$ and $\vec{OC}_t$ represents the vector from the origin o to the semantic center c_t . To learn semantic information between different images, we divide each batch size into two parts, and sequentially obtain images x_i and x_j . By calculating the cosine similarity between the mapped features $A_{i,t}$ and $A_{j,t}$ at the same positions in different images x_i and x_j , we can obtain the similarity measures at different locations.

$$\text{sim}(\vec{C}_t A_{i,t}, \vec{C}_t A_{j,t}) = 1 - \frac{\vec{C}_t A_{i,t} \cdot \vec{C}_t A_{j,t}}{\|\vec{C}_t A_{i,t}\| \cdot \|\vec{C}_t A_{j,t}\|} \quad (9)$$

The subsequent use of $\text{sim}_{ij,t}$ is to represent $\text{sim}(\vec{C}_t A_{i,t}, \vec{C}_t A_{j,t})$. When $\text{sim}_{ij,t}$ is smaller, it indicates a higher semantic similarity between $A_{i,t}$ and $A_{j,t}$. We selectively reduce the distance between the H most semantically similar mapped features, i.e., decreasing their similarity measure. The similarity measure between the H most similar mapped features is expressed as $\left\{ \text{sim}_{ij,h} \mid h \in \arg \min_H (\text{sim}_{ij,t}) \right\}$. $L_{\text{inter-att}}$ is defined as the penalty based on $\text{sim}_{ij,h}$:

$$L_{\text{inter-att}} = \frac{1}{H} \sum_{h=1}^H \text{sim}_{ij,h} \quad (10)$$

2) Inter-Constraints: As shown in the Fig. 6, relying solely on inter-attraction could lead to an incorrect optimization tendency during the mapping process, where correctly mapped features are wrongly attracted to semantic centers that are not semantically similar. We refer to this phenomenon as erroneous optimization. Inspired by previous research [42], which suggests preventing mismatches by limiting the distance between the most similar matching pairs, we establish inter-constraints by limiting the distances from $\phi(F_{i,t})$ and $\phi(F_{j,t})$ to their respective G most similar mapped features in the image. As shown in the Fig. 6, through inter-constraints, the targets of inter-attraction, $\phi(F_{i,t})$ and $\phi(F_{j,t})$, can adjust their positions based on the distance to neighboring features, allowing the inter-attraction to guide the feature mapping in a more accurate direction and avoid erroneous

optimization. The distance to the G nearest mapped features $\phi(F_{i,r})$ for the mapped feature $\phi(F_{i,t})$ is represented as $\left\{ d_{i,t}^g \mid g \in \arg \min_G (\phi(F_{i,t}), \phi(F_{i,r}), t \neq r) \right\}$. The $L_{\text{inter-con}}$ is set to prevent the distances $d_{i,t}^g$ from being extended:

$$L_{\text{inter-con}} = \max(d_{i,t}^g) \quad (11)$$

Therefore, the combined use of intra- and inter-correlation learning for transfer learning to optimize the parameters of $\phi(\cdot)$ is defined by the total loss as follows:

$$\text{Loss} = L_{\text{intra-att}} + L_{\text{intra-rep}} + L_{\text{inter-att}} + L_{\text{inter-con}} \quad (12)$$

E. Anomaly Score

Inspired by [13] and [26], we adopt the Euclidean distance between the K nearest semantic centers and mapped features $\phi(F_{i,t})$ to calculate the anomaly score of mapped features, as follows:

$$D_{i,t,k} = \min_k D(\phi(F_{i,t}), c_t) \quad (13)$$

The closer the mapped feature $\phi(F_{i,t})$ is to the semantic center c_t^k , the more evident the normal feature is. We employ the *softmax* function to calculate the weights of the feature $\phi(F_{i,t})$ with respect to these K centers.

$$S_{i,t} = \frac{e^{-D_{i,t,0}}}{\sum_{k=1}^K e^{-D_{i,t,k}}} \cdot D_{i,t,0} \quad (14)$$

where $S_{i,t}$ represents the anomaly score of the mapped feature $\phi(F_{i,t})$. We use the maximum feature anomaly value $S_{i,t}$ in the image as the image's anomaly value $\text{Sim}_{\text{age}_i} = \max(S_{i,t})$. By calculating the *Precision* and *Recall* between each image's anomaly value $\text{Sim}_{\text{age}_i}$ and its label y_i , we determine the *F1*. Finally, we use the *F1* of the image to determine the threshold for the image anomaly score, as follows:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (15)$$

$$\text{Score}_i = \begin{cases} 1, & \text{Sim}_{\text{age}_i} \geq \arg \max(F1) \\ 0, & \text{Sim}_{\text{age}_i} < \arg \max(F1) \end{cases} \quad (16)$$

where $\arg \max(\cdot)$ represents the function that returns the index of the maximum value in the *F1* array.

IV. EXPERIMENTS

A. Public Benchmarks

We utilized three publicly available chest X-ray image datasets to evaluate the performance of the FDS, including: the ZhangLab dataset for pediatric pneumonia, the RSNA dataset for adult pneumonia, and the CheXpert dataset covering 12 complex diseases. These datasets were developed and curated by their respective authors and included ethical statements for research purposes, ensuring the absence of potential biases.

The **ZhangLab Chest X-ray** dataset [43] was released by the Shiley Eye Institute, University of California San Diego,

TABLE I

BENCHMARK TESTING RESULTS FOR THE TESTING DATASETS. THE INDICATORS FOR THE TOP THREE POSITIONS IN THE BASELINE ARE REPRESENTED BY **ORANGE**, **BLUE**, AND **GREEN**, RESPECTIVELY. T-S REFERS TO THE TEACHER-STUDENT ARCHITECTURE

Type	Method	Year	ZhangLab			RSNA			CheXpert		
			ROCAUC	F1	ACC	ROCAUC	F1	ACC	ROCAUC	F1	ACC
Anomaly Simulation	PII	2021	0.871	0.866	0.830	0.775	0.707	0.728	0.751	0.682	0.686
	CutPaste	2021	0.681	0.747	0.667	0.571	0.517	0.567	0.581	0.529	0.584
	FPI	2022	0.786	0.819	0.753	0.541	0.441	0.560	0.740	0.719	0.700
	NSA	2022	0.843	0.833	0.772	0.766	0.660	0.697	0.664	0.610	0.658
	Simplenet	2023	0.807	0.833	0.784	0.680	0.624	0.650	0.626	0.681	0.600
	AnatPaste	2023	0.901	0.860	0.829	0.775	0.746	0.730	0.817	0.753	0.760
	CropMixPaste	2023	0.868	0.856	0.803	0.659	0.619	0.635	0.770	0.700	0.710
T-S	RD	2022	0.727	0.813	0.737	0.556	0.667	0.533	0.608	0.672	0.512
	RD++	2023	0.913	0.882	0.859	0.666	0.633	0.632	0.654	0.622	0.624
	EDC	2023	0.875	0.863	0.821	0.672	0.593	0.652	0.685	0.649	0.648
	SQUID	2023	0.848	0.811	0.776	0.649	0.618	0.632	0.780	0.755	0.726
Distribution Mapping	Padim	2021	0.835	0.850	0.801	0.633	0.659	0.533	0.617	0.665	0.572
	Fastflow	2021	0.907	0.865	0.827	0.654	0.674	0.580	0.646	0.667	0.576
	Csflow	2022	0.838	0.855	0.798	0.761	0.713	0.643	0.778	0.735	0.680
	CFA	2022	0.895	0.854	0.821	0.768	0.717	0.718	0.748	0.698	0.708
	PatchCore	2022	0.817	0.806	0.752	0.729	0.678	0.690	0.720	0.637	0.672
	Mean-shifted	2023	0.938	0.905	0.878	0.759	0.702	0.708	0.783	0.692	0.726
	Pyramidflow	2023	0.817	0.816	0.766	0.622	0.623	0.610	0.608	0.616	0.592
	Ours		0.979	0.967	0.959	0.829	0.755	0.757	0.861	0.791	0.798

USA. It comprises 5,232 pediatric chest X-ray images. The training dataset comprises 3,883 images depicting pneumonia cases and 1,349 images depicting normal cases. The testing dataset includes 234 normal images and 390 pneumonia images from a total of 624 patients. Based on the settings described in [12], we completed the model training on the training dataset. To prevent the leakage of information from the testing dataset, we tested using the weights obtained in the last epoch of the training phase.

The **RSNA** dataset [44] released by the Clinical Center of the National Institutes of Health, USA, includes chest X-ray images from 26,684 patients. Among these images, 8,851 are marked as normal, 6,012 show pneumonia, and another 11,821 are neither normal nor indicative of pneumonia. Following the setup in [34], we train our models on the training dataset and use the model weights from the final training epoch for testing.

The **Stanford CheXpert** dataset [45] was released by the Department of Radiology, Stanford University, USA. It encompasses 12 diseases, namely Pleural Effusion, Enlarged Cardiomeastinum, Lung Opacity, Consolidation, Pleural Other, Fracture, Atelectasis, Lung Lesion, Cardiomegaly, Edema, Pneumonia, and Pneumothorax. Our evaluation was limited to anterior-posterior (PA) view images, comprising 5,249 normal chest X-ray images and 23,671 chest X-ray images exhibiting diseases. Following the setup in [12], we trained on the training dataset and tested using the weights from the last epoch.

B. Experimental Settings

The experiments in this paper were conducted using Python 3.9 and PyTorch 1.12.1 on the Ubuntu system, equipped with an Intel i5-12490F CPU and an NVIDIA GEFORCE RTX 3090 24G GPU. We chose to use the WideResNet50 pre-trained on ImageNet as the backbone network for the initial experiments and the WideResNet50 backbone network was fully frozen. We used 1×1 CoordConv as the feature

mapper and optimized it during the training phase. In the preprocessing phase, the size of each chest X-ray image was resized to 224×224 pixels, the same size as the images pre-trained on ImageNet. During training, we used AdamW as the optimizer, set the batch size to 8, and the training period to 100 epochs. The learning rate was set to 5×10^{-4} , and the weight decay was set to 3×10^{-5} . Finally, in the intra-correlation learning, the hyperparameters K and J were both set to 3. The initial value of the radius R was set to 1×10^{-5} , and gradient descent was allowed. In the inter-correlation learning, the hyperparameters H were set to 3, and G was set to 1. All hyperparameter selections were based on good experimental performance achieved on the selected network (WideResNet50).

C. Comparison With State-of-the-Art

We employed 18 advanced baseline models for training and testing on the ZhangLab Chest X-ray and Stanford CheXpert and RSNA datasets, which encompass methods such as anomaly simulation, Teacher-Student architecture, and distribution mapping. They are: PatchCore [25], Padim [24], RD [17], RD++ [20], Csflow [21], Fastflow [22], Pyramidflow [36], Simplenet [15], CFA [26], NSA [9], PII [8], FPI [7], CutPaste [33], CropMixPaste [10], SQUID [12], EDC [11], AnatPaste [34], Mean-Shifted [27]. We evaluated their performance using standard metrics: Area Under the ROC Curve (ROCAUC), Accuracy (ACC), and F1 Score (F1). The performance of our proposed FDS and various state-of-the-art baseline models is presented in Table I.

1) **ZhangLab Chest X-Ray**: We attained superior results compared to other baselines, achieving ROCAUC, F1, and ACC of 0.979, 0.967, and 0.959, respectively. The methods PII and CropMixPaste, employing **Anomaly Simulation** techniques, yielded favorable results due to their effective anomaly simulation strategies, achieving ROCAUC of 0.871 and 0.868, respectively. Based on this, AnatPaste further enhances the

process by using a threshold-based lung segmentation pre-processing task to directly create pseudo-anomalies in the thoracic area of X-ray images, achieving ROCAUC of 0.901. The **Teacher-Student** Architecture method RD++, employed on the ZhangLab dataset, which exclusively contains pneumonia anomalies, achieved excellent results through compact feature representation and the suppression of abnormal features, with ROCAUC, ACC, and F1 of 0.913, 0.882, and 0.859, respectively. Among **Distribution Mapping**-based methods, Fastflow demonstrated outstanding results owing to its Transformer-based architecture’s ability to better attend to anatomical structures in X-ray images. Additionally, CFA utilizing proposed coupled hyperspherical adaptation, enhances feature distinctiveness. Mean-Shifted achieved ROCAUC of 0.938 through its contrastive loss.

2) **RSNA**: The dataset includes imaging related to adult pulmonary diseases, primarily focusing on pneumonia and its complications. In comparison with other benchmark models, our method achieved ROCAUC, F1, and ACC values of 0.829, 0.755, and 0.757, respectively. In the **anomaly simulation** methods, PII and AnatPaste, which simulate anomalies based on the natural changes in normal anatomical structures, both achieved good ROCAUC results of 0.775. EDC based on a **Teacher-Student** architecture, effectively and consistently learned the features of X-ray images through the introduction of a contrastive learning paradigm. In the **distribution mapping** methods, Fastflow primarily learns data distributions through bijective transformations, focusing heavily on local texture features, which resulted in poor performance in global semantic understanding and thus did not meet expectations in the RSNA dataset. Meanwhile, CFA maintained good performance in processing images of adult lung turbidity through its adaptive strategy.

3) **Stanford CheXpert**: We achieved superior results compared to other baselines, with ROCAUC, F1, and ACC reaching 0.861, 0.791, and 0.798, respectively. The **Anomaly Simulation** method proposed by CropMixPaste tends to generate lesions resembling density shadows, leading to improved performance on the more complex CheXpert dataset, which includes 12 diseases. Meanwhile, AnatPaste specifically designed for simulating pneumonia lesions, also demonstrates good performance on the CheXpert dataset. The **Teacher-Student** Architecture method RD++ demonstrated superior performance on the ZhangLab dataset. However, its performance is not as good on the more complex CheXpert dataset compared to our method, indicating that complex medical lesions affect the effectiveness of RD++’s compact feature representation and suppression of abnormal features. SQUID utilizing spatial-aware memory queues to detect lesions, exhibits a better focus on lesion areas in more complex chest X-ray images. In the **Distribution Mapping**-based methods, Csflo achieves excellent results on complex datasets through joint likelihood estimation of multiscale features, achieving ROCAUC of 0.778.

D. Ablation Studies

1) **Impact of Different Modules**: Table II presents the results of an ablation study assessing the contributions of different

TABLE II

WE EVALUATED THE PERFORMANCE OF MODELS WITH DIFFERENT COMPONENTS ON THE ZHANGLAB CHEXPert AND RSNA DATASETS. RED INDICATES THE DECREASE IN INDICATORS DUE TO THE LACK OF INTER-CONSTRAINTS

Dataset	AFPF	Intra		Inter		ROCAUC	ACC	F1
		Att	Rep	Att	Con			
ZhangLab						0.889	0.819	0.859
	✓	✓				0.947	0.896	0.917
	✓	✓	✓			0.971	0.954	0.964
	✓	✓	✓	✓		0.978	0.952	0.962
	✓	✓	✓	✓	✓	0.979	0.959	0.967
CheXpert		✓				0.699	0.652	0.656
	✓	✓				0.722	0.680	0.689
	✓	✓	✓			0.785	0.740	0.710
	✓	✓	✓	✓		0.848	0.776	0.776
	✓	✓	✓	✓	✓	0.861	0.798	0.791
RSNA		✓				0.742	0.697	0.647
	✓	✓				0.751	0.707	0.711
	✓	✓	✓			0.797	0.739	0.742
	✓	✓	✓	✓		0.815	0.740	0.741
	✓	✓	✓	✓	✓	0.829	0.757	0.755

TABLE III

ON THE ZHANGLAB, CHEXPert, AND RSNA DATASETS, WE EXAMINED THE IMPACT OF DIFFERENT ATTENTION MODULES ON OUR AFPF MODULE

Method	ZhangLab ROCAUC	CheXpert ROCAUC	RSNA ROCAUC
Coordinate	0.876	0.710	0.726
SA	0.868	0.648	0.690
HaloNet	0.856	0.645	0.707
ECA	0.874	0.661	0.677
Triplet	0.881	0.678	0.705
SK	0.907	0.703	0.719
Ours	0.979	0.861	0.829

components of FDS. Initially, using only intra-attraction for transfer learning to address correct mapping scenarios yielded ROCAUC scores of 0.889, 0.699, and 0.742 on the ZhangLab, CheXpert, and RSNA datasets, respectively. Utilizing the AFPF module allows the retention of local details while effectively capturing subtle lesions in the chest area, significantly influencing the ZhangLab dataset, which contains only pneumonia abnormalities, achieving ROCAUC of 0.947. Incorporating intra-repulsion effectively manages fuzzy mapping, achieving ROCAUC scores of 0.971, 0.785, and 0.797 on the ZhangLab, CheXpert, and RSNA datasets, respectively. When only inter-attraction is used to address error mappings, the ROCAUC on the ZhangLab and RSNA datasets reaches 0.978 and 0.815, respectively. However, due to some correct mappings lacking inter-constraints and being mistakenly optimized as error mappings, the ACC and F1 scores on the ZhangLab dataset decrease by 0.002 each, and the F1 score on the RSNA dataset decreases by 0.001. By introducing inter-constraints, optimal results are achieved on the ZhangLab, CheXpert, and RSNA datasets, with ROCAUC values of 0.979, 0.861, and 0.829, respectively.

2) **Impact of Attention**: In Table III, we evaluated the performance of various attention mechanisms in the feature fusion module. Coordinate [46] embeds positional information into channel attention, capturing long-range dependencies while preserving precise location information. Shuffle Attention (SA) [47] optimizes information processing by combining spatial and channel attentions with Shuffle units and channel shuffle operations. HaloNet [48] enhances accuracy by introducing

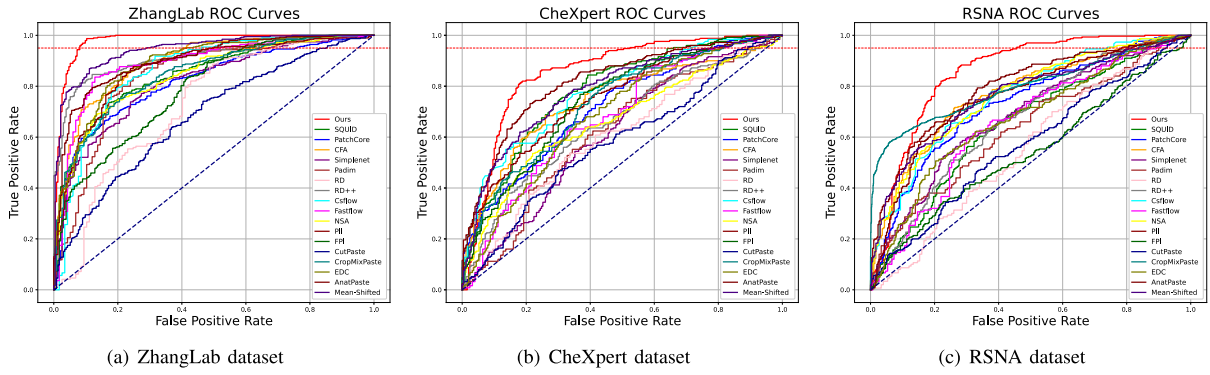


Fig. 7. ROC Curves for the ZhangLab, CheXpert, and RSNA datasets are presented. When the TPR (True Positive Rate) reaches 0.95, the intersection of the red line with the ROC Curves indicates the corresponding FPR (False Positive Rate), referred to as FPR@TPR95.

chunked local attention and attention downsampling techniques. ECA [49] effectively captures local cross-channel interactions using one-dimensional convolutions. Triplet [50] effectively captures cross-dimensional interactions through a three-branch structure. SK [51] enhances the neural network’s adaptability by introducing a dynamic selection mechanism among multiple kernel sizes. Upon comparisons, we found that the Multihead-attention module is more suitable for our Anatomical Feature Pyramid Fusion (AFPF) and outperforms other modules in performance. Therefore, we believe that after employing the Multihead-attention module, it can better capture the anatomical structures in X-ray images.

3) *Impact of Hyperparameters*: We investigated the impact of four hyperparameters, namely K, J, H, and G, within the intra- and inter-correlation learning Top-k setting of FDS. As illustrated in Table IV, the best results were obtained when K, J, and H were all set to 3, with model performance declining as these values increased. The optimal result was achieved when G was set to 1; however, Setting G to 2 or 3 resulted in a notable decline in model performance due to increased constraints, which restricted intra-attraction and consequently led to a significant decrease in performance.

4) *ROC Curve*: As shown in Fig. 7, we have demonstrated the image AUC curves of the FDS method on the ZhangLab, CheXpert, and RSNA datasets. The key metric (FPR@TPR95) is the false positive rate (FPR) when the true positive rate (TPR) reaches 0.95, marked with a red line. FPR@TPR95 is the proportion of healthy samples misdiagnosed as diseased when nearly all true cases are detected, which is crucial for medical judgment in healthcare applications. The more leftward the intersection of the AUC curve with the standard red line (TPR=0.95), the lower the FPR@TPR95, indicating a lower rate of misdiagnosis. In these three chest X-ray image datasets, the FDS method achieved the lowest FPR@TPR95, demonstrating its potential high accuracy in the field of medical anomaly detection. This series of quantitative analyses further validates the FDS method’s exceptional anomaly detection performance across different styles of chest X-ray image datasets.

E. Robustness Experiment

1) *Whether Pre-Trained Bias Is Mitigated?*: Due to biases introduced by using an ImageNet pre-trained network for

TABLE IV

THE SETTING OF THE TOP-K HYPERPARAMETERS K, J, H, G IN INTRA- AND INTER-CORRELATION LEARNING HAVE AN IMPACT ON THE FDS

Hyperparameters (K,J,H,G)	ZhangLab ROCAUC	ZhangLab ACC	ZhangLab F1
(1,3,3,1)	0.973	0.947	0.959
(3,3,3,1)	0.979	0.959	0.967
(5,3,3,1)	0.972	0.947	0.959
(7,3,3,1)	0.961	0.928	0.943
(10,3,3,1)	0.942	0.896	0.920
(20,3,3,1)	0.816	0.758	0.818
(3,1,3,1)	0.975	0.949	0.960
(3,3,3,1)	0.979	0.959	0.967
(3,5,3,1)	0.977	0.954	0.963
(3,7,3,1)	0.975	0.945	0.958
(3,10,3,1)	0.963	0.924	0.937
(3,20,3,1)	0.874	0.829	0.871
(3,3,1,1)	0.978	0.954	0.963
(3,3,3,1)	0.979	0.959	0.967
(3,3,5,1)	0.976	0.944	0.956
(3,3,7,1)	0.972	0.942	0.955
(3,3,10,1)	0.955	0.919	0.933
(3,3,20,1)	0.850	0.816	0.851
(3,3,3,1)	0.979	0.959	0.967
(3,3,3,2)	0.770	0.745	0.703
(3,3,3,3)	0.753	0.704	0.779

feature extraction, we conducted ablation studies with a model pre-trained on medical images to explore whether the FDS can mitigate these biases.

ChexNet [52] provides DenseNet121 weights (called ChexDense) pre-trained on 112,120 frontal chest X-ray images involving 30,805 unique patients with 14 disease labels.

RadImageNet [53] provides DenseNet121 weights (called RadDense) and ResNet50 weights (called RadRes) pre-trained on 1.35 million annotated images of musculoskeletal, nervous system, oncology, gastrointestinal, endocrine, and pulmonary pathologies from CT, MRI, and ultrasound images. These images include medical images marked with 3 modalities, 11 anatomical structures, and 165 pathology labels.

As shown in Fig. 8(a), the model pre-trained on ImageNet achieved ROCAUC of only 0.441 in the first epoch, which was due to significant bias from the pre-trained weights. In contrast, the ChexDense weights, pre-trained on medical data, reached ROCAUC of 0.676, while the RadDense weights achieved ROCAUC of 0.722. As training progressed, it was observed that FDS could reduce the bias from using ImageNet pre-trained weights, showing significant improvement by the second epoch and surpassing the performance of models using

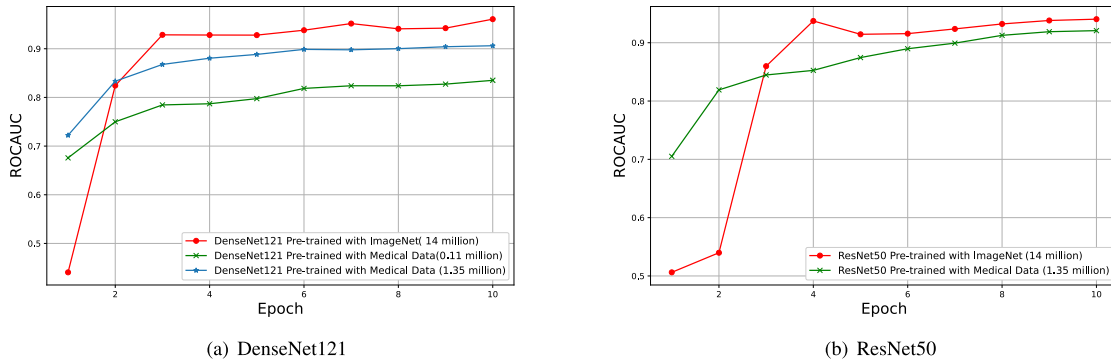


Fig. 8. Training processes of (a) DenseNet121 and (b) ResNet50 networks pre-trained on various datasets.

medical pre-trained weights by the third epoch, maintaining higher performance in subsequent training. Additionally, Fig. 8(a) also demonstrates that models pre-trained on larger medical datasets can achieve better detection performance, although their performance still did not exceed that of the models pre-trained on ImageNet, because the large data volume of ImageNet provides richer and more discriminative features for the model, which is why we continued to use the ImageNet pre-trained model as the feature extractor.

Similarly, we compared the performance on ResNet50, as shown in Fig. 8(b). The model pre-trained on ImageNet achieved ROCAUC of 0.506 in the first epoch, indicating an initial bias caused by the pre-trained ImageNet data. The RadRes weights achieved ROCAUC of 0.705 in the first epoch, showing a better initial performance. As training progressed, it was demonstrated that FDS could effectively alleviate the bias introduced by the ImageNet pre-trained data, enhancing model performance.

2) Disease-Free Training Requirement?: In medical anomaly detection, so-called unsupervised methods are not fully unsupervised, as the model requires training on healthy images. The removal of images without lesions depends on manual annotation of healthy or diseased labels at the image space. Consequently, there is a phenomenon where abnormal samples are erroneously labeled. To assess the robustness of the model, we evaluated our approach against four other distribution mapping paradigms within the ratios (disease images/ healthy images) of 0% to 40% proportions. As shown in Fig. 9, when the ratio is between 0% and 4%, FDS experiences a significant decline from 0.979 to 0.837. This is because FDS learns abnormal patterns through intra-attraction that we do not want to learn. However, as the ratio gradually increases from 4% to 40%, FDS does not exhibit a continued severe decline as anticipated. This is because intra-correlation better ignores some abnormal patterns due to its handling of fuzzy mapping, while the constraints of inter-correlation learning ensure a better optimization direction for error mapping. Conversely, as the ratio escalates, CFA consistently shows a downward trend. Padim, which models the normal class probabilities using a multivariate Gaussian distribution, undergoes a gradual decrease after experiencing slight fluctuations between 8%-12%. On the other hand, PatchCore, employing core subset sampling to reduce redundancy while potentially missing some abnormal patterns, exhibits a variability range from 0% to 40%.

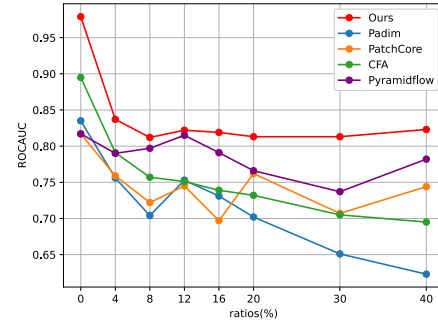
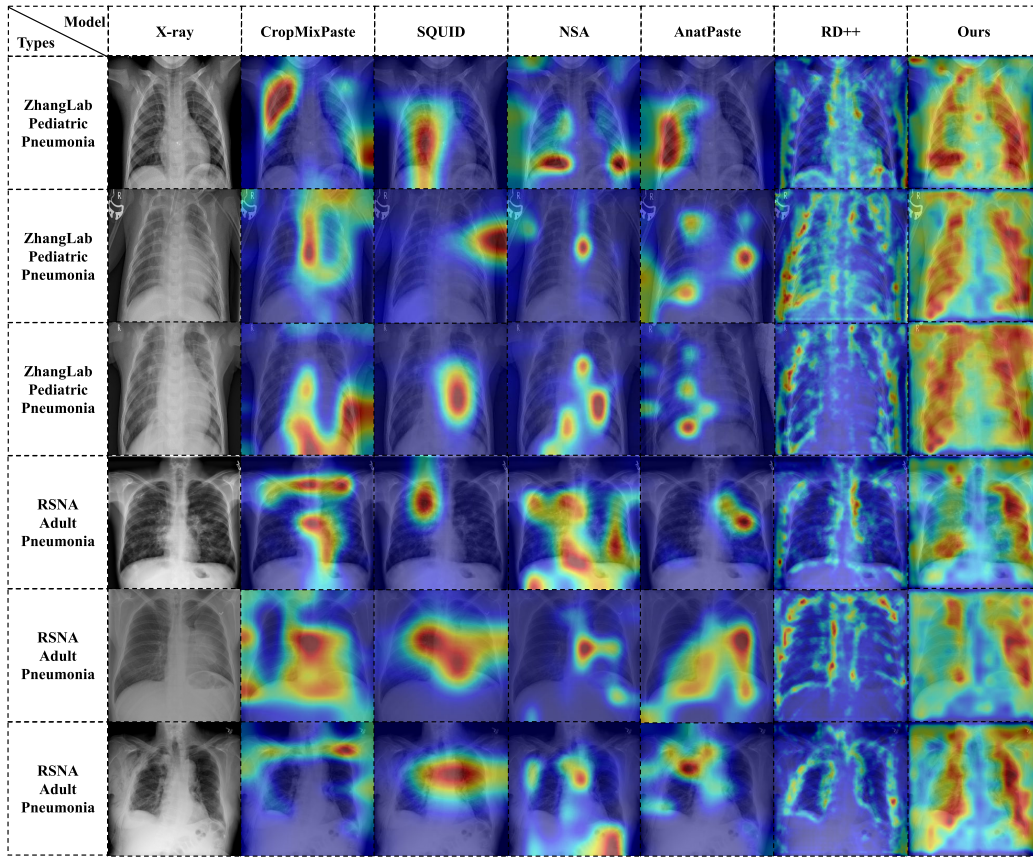


Fig. 9. Several advanced models of the distribution mapping type undergoing robustness analysis.

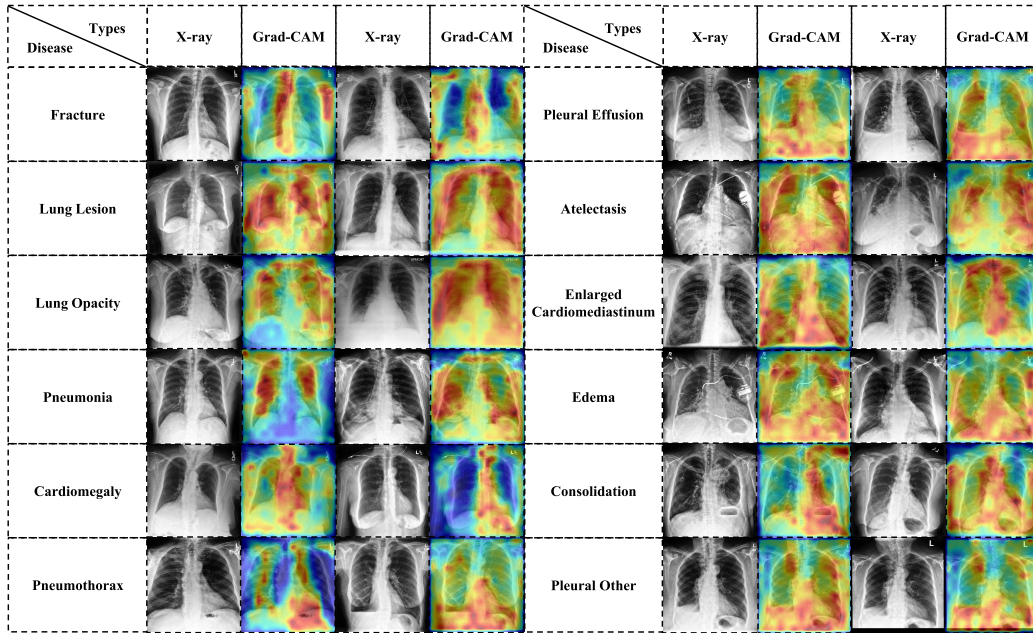
F. Visualization Results

Grad-CAM has been employed for anomaly localization [10], [25], [33]. Given the similarity between normal and abnormal textures, obtaining pixel-level labels for localized anomalies in chest X-ray images can be challenging. Therefore, we utilize Grad-CAM visualization [54] to identify regions most likely to exhibit anomalies. In Fig. 10, we present the visual comparison results of pneumonia X-ray images (from the pediatric pneumonia ZhangLab dataset and the adult pneumonia RSNA dataset) and multi-disease chest X-ray images (from the CheXpert dataset). We utilize features from the final convolutional layer of each model to compute gradients for Grad-CAM.

1) Pneumonia X-Ray: In Fig. 10(a), we showcase the localization capabilities of several advanced algorithms for pediatric pneumonia (ZhangLab) and adult pneumonia (RSNA). FDS demonstrates a strong ability to focus on the entire lung, effectively covering the entire lesion area. Through the lung segmentation preprocessing task, AnatPaste can effectively simulate and focus on local lesions in the lungs; however, it may overlook anomalies in the diffuse interstitial pattern. The method proposed by CropMixPaste mainly concentrates on specific local areas, which may lead to the neglect of the entire lesion. Additionally, since its anomaly simulation is not limited to the lungs, it may cause attention to be diverted to areas outside the lungs. NSA seems to focus on overly subtle areas, which may make representation learning challenging as it pursues naturally simulated anomalies. In complex chest X-ray images, RD++ utilizes a student network to learn normal feature distributions, with the learned content tending toward the corners of chest X-ray images. SQUID using spatial-aware



(a) Pediatric and Adult Pneumonia X-rays



(b) Multi-disease X-rays

Fig. 10. (a) Visual results of Grad-CAM for pneumonia using CropMixPaste, SQUID, NSA, AnatPaste, RD++, and our proposed method. (b) Visual results of Grad-CAM using our proposed method on the CheXpert dataset for twelve different diseases.

memory queues for anomaly detection, can effectively focus on the lungs. However, due to automatically omitting patterns of minority groups, when medical instruments are present, it may lead to attention being concentrated on areas we do not wish to emphasize.

2) Multi-Disease X-Ray: Given the diversity of chest diseases, in Fig. 10(b), we further present the Grad-CAM visualization results of FDS on the CheXpert dataset, which comprises 12 diseases. We observe that FDS currently struggles to precisely locate anomalies at the pixel level. However,

this limitation is understandable for several reasons. Firstly, FDS operates as an unsupervised approach, unlike methods such as [1], [2], [3], [4], and [6], which rely on pixel-level annotations for normal and abnormal images. Secondly, the diverse nature of chest diseases can lead to various complications. For instance, pneumonia may cause fluid accumulation between the lung and the pleura (the thin tissue layer surrounding the chest cavity), resulting in pleural effusion. Consequently, the complexity of chest X-ray images further hampers pixel-level localization. Nonetheless, FDS demonstrates the ability to comprehensively cover lesion areas for the 12 diseases provided by the CheXpert dataset, thereby effectively aiding medical experts in diagnosis.

V. DISCUSSION

Based on our experimental results, our method outperforms existing methods on the pediatric pneumonia ZhangLab dataset, the adult pneumonia RSNA dataset, and the multi-disease CheXpert dataset. In methods based on anomaly simulation, the significant differences between simulated pseudo-lesions and real lesions in the test datasets greatly limit the performance of such models. In methods based on teacher-student networks, due to the subtlety of the lesions, the student network tends to generalize and generate abnormal feature distributions similar to those of the teacher network. Methods based on distribution mapping inevitably use pre-trained networks as feature extractors. Although Pyramidflow attempts to avoid using the pre-trained network through defect contrast localization, the results are not ideal. Existing distribution mapping-based methods mainly rely on similarity measures (e.g., Euclidean distance, Mahalanobis distance, cosine similarity) for transfer learning and anomaly detection. For instance, CFA and PatchCore use Euclidean distance as a similarity measure for training. This neglects the similarity differences between different dataset domains, thereby affecting the model's feature representation capability. Mean-shifted through an improved contrastive loss, brings the positive samples closer together, allowing the model to indirectly learn the similarity of the target X-ray dataset, thus achieving a good detection performance. However, the data augmentation methods Mean-shifted employs might introduce anomalies (e.g., Flipping X-ray images may cause abnormal phenomena such as situs inversus), thereby hindering detection performance. Therefore, we address the similarity differences by distinguishing between possible mappings (correct mapping, fuzzy mapping, erroneous mapping) and using intra-correlation and inter-correlation learning. Our method avoids the risk of introducing anomalies, while approximating the similarity of the target domain, thereby enhancing the model's feature representation capability.

Although our method FDS, demonstrates excellent anomaly detection performance across several datasets, it confronts a common challenge faced by most X-ray anomaly detection methods: accurately localizing lesions. This difficulty stems from the presence of tissues with varying densities on the same plane in X-ray images, which creates only subtle differences between lesions and normal tissues.

VI. CONCLUSION

In this work, we propose an intra- and inter-correlation framework FDS for exploring anomaly detection in chest X-ray images. Firstly, our FDS introduces an AFPF module that can fully utilize the anatomical structural information in chest X-ray images. Subsequently, we employ an intra- and inter-correlation learning strategy for facing the differences of similarity introduced by the pre-trained network. With these techniques, the method can alleviate the challenge of anomaly detection in diverse chest disease environments. To assess the effectiveness of our approach, experiments were conducted on two different types of datasets. The experimental results demonstrate that our method surpasses several state-of-the-art methods in chest X-ray anomaly detection.

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